

The YouTube Collaboration Network: Structural Patterns and Popularity Dynamics

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Project Report

Introduction

This project addresses the problem of understanding how content creators on YouTube collaborate across genres, languages, and networks, and how these collaborations influence popularity, and engagement. We implement network analysis techniques on a youtube collaboration dataset, which represents collaborations between YouTubers as a directed, weighted graph. The need for this project arises from the lack of accessible, large-scale analysis of content-based collaboration patterns and their impact on visibility and community structure. By conducting this study, we aim to provide insights into genre-based clustering, language effects, network centrality, and influencer identification in online media ecosystems.

Dataset

The dataset has been taken from the results of the paper, **Collaborations on YouTube: From Unsupervised Detection to the Impact on Video and Channel Popularity**, C. Koch, M. Lode, D. Stohr, A. in which 2.4 years of video content and 3 months of youtube channel stats have been scrapped. The collaborations are detected using CATANA, a face recognition system on the basis of co-appearance in videos.

Nodes	Edges
<pre>{ "id": 0, "label": " UCm1Lj07mpzb68Q71PjGleWQ", "cluster": "1621", "network": "None" }</pre>	<pre>{ "source": 3619, "target": 464, "cluster": "5601", "weight": 1, "videos": "lAgzb01QRtA" }</pre>

More file details are as follows: The direction here is from guest youtuber to host youtuber. The number of collaborations is used as edge weights.

- Tracks channel metrics (views, subscribers, comments, collaborations).
- Language data for channels.
- Direct creator collaborations.
- Video-level views and collaborations.
- Video and channel mentions.
- Graph of collaborations.

1 Degree-Degree Correlation Analysis

To analyze connectivity patterns among YouTube channels in our collaboration network, we examine the **degree-degree correlation**. This assesses whether high-degree nodes (e.g., popular creators) tend to connect with other high-degree nodes, or with low-degree ones.

1.1 Approach

We constructed a directed collaboration graph from the dataset and converted it into an undirected form to compute the Pearson degree assortativity coefficient. This coefficient measures the tendency of nodes to link with similar-degree nodes:

- A positive value indicates assortative mixing — nodes connect with others of similar degree.
- A negative value implies disassortative mixing — high-degree nodes connect more with low-degree nodes.
- A value close to zero suggests no strong correlation in degree between connected nodes.

To visualize this, we created a scatter plot showing the degree of source and target nodes for all edges in the undirected network.

1.2 Degree-Degree Scatter Plot

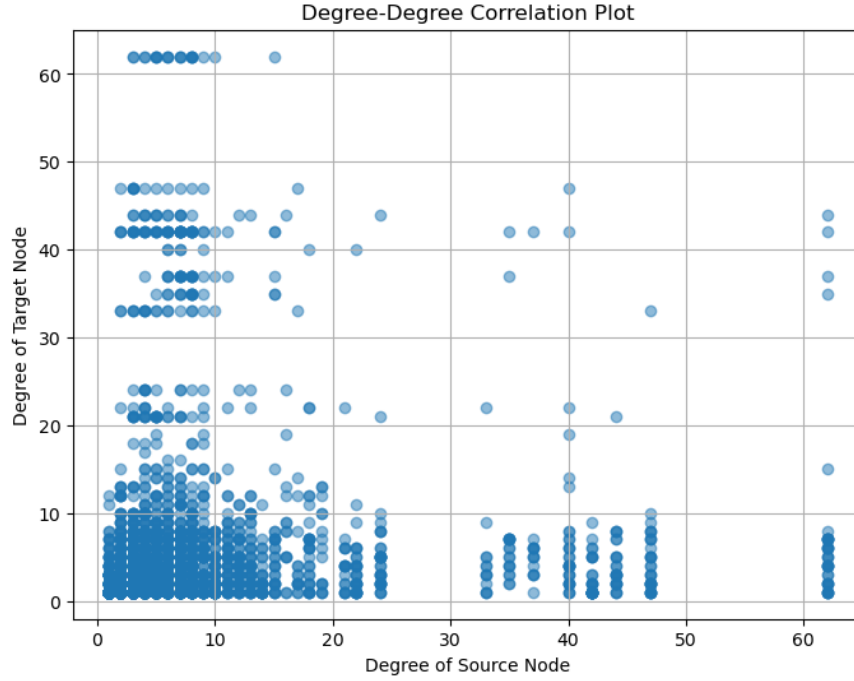


Figure 1: Degree-Degree Correlation Plot: Each point represents a pair of connected nodes, with their respective degrees.

1.3 Observations

- The scatter plot reveals a dense concentration of points in the region where both source and target node degrees are between 1 and 15. This indicates that most collaborations happen among small to moderately connected YouTube channels.
- There is a noticeable presence of vertical streaks where a few source nodes (with degrees above 40 or 60) connect to many target nodes with a broad range of degrees. These are indicative of **hub nodes** — highly influential channels collaborating with both large and small creators.
- Conversely, there are horizontal streaks formed by low-degree source nodes (e.g., degree < 10) connecting with target nodes of very high degree (even up to 60+), further emphasizing the presence of **star collaborators or influencers** who attract a large number of smaller creators.
- The absence of a strong diagonal trend in the plot (i.e., no clustering along the $y = x$ line) suggests that the network does not exhibit assortative mixing — creators do not preferentially collaborate with others of similar popularity (as measured by degree).

2 Degree distribution

The dataset under study represents a YouTube collaboration network extracted from a ‘.gml’ file, where each node corresponds to a distinct YouTube channel and each edge signifies a collaboration between two channels. An edge weight represents the number of videos in which the pair has collaborated.

- **In-degree:** Most channels are invited by very few others, indicating limited inbound collaborations. A long-tail distribution suggests a few channels are frequently invited, indicating popularity or influence.
- **Out-degree:** Similarly, most channels initiate collaborations with very few others. However, a handful of nodes demonstrate high out-degrees, marking them as collaboration hubs or initiators.
- **Edge Weights:** The majority of channel pairs collaborate only once, whereas some exhibit repeated collaborations, indicating strong or ongoing partnerships.
- **Collaboration Count:** The total degree (in + out) illustrates overall engagement in the network. Most nodes show limited activity, while some are highly central.

2.1 Distributions and Observations

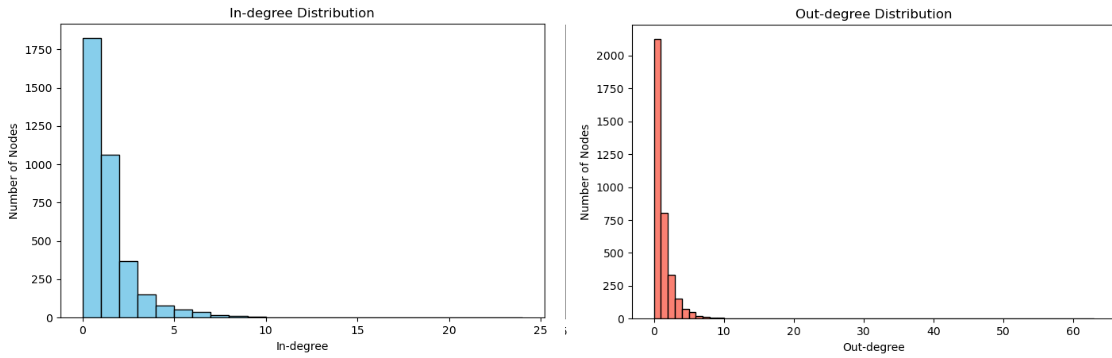


Figure 2: In-degree and Out-degree Distributions

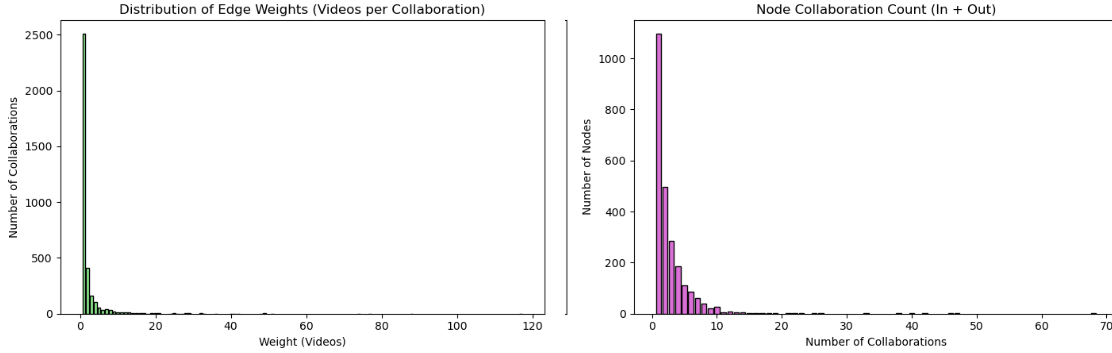


Figure 3: Edge Weight and Total Collaboration Count Distributions

These plots reveal that the network follows a typical **scale-free behavior**, where a few nodes dominate in terms of connectivity. This pattern is indicative of influencer-heavy structures commonly observed in social and creator networks.

3 Analysis of Collaboration Patterns across Popularity, Topic, Category, and Language

To explore collaboration dynamics among YouTube creators, we analyzed collaboration pair counts based on four key dimensions: **popularity level**, **content topic**, **channel category**, and **channel**

language. The analysis is based on an enriched edge list from the file `df_collabs.txt`, with metadata such as popularity, topic, category, and language labels for both collaborating channels.

3.1 Popularity-Based Collaborations

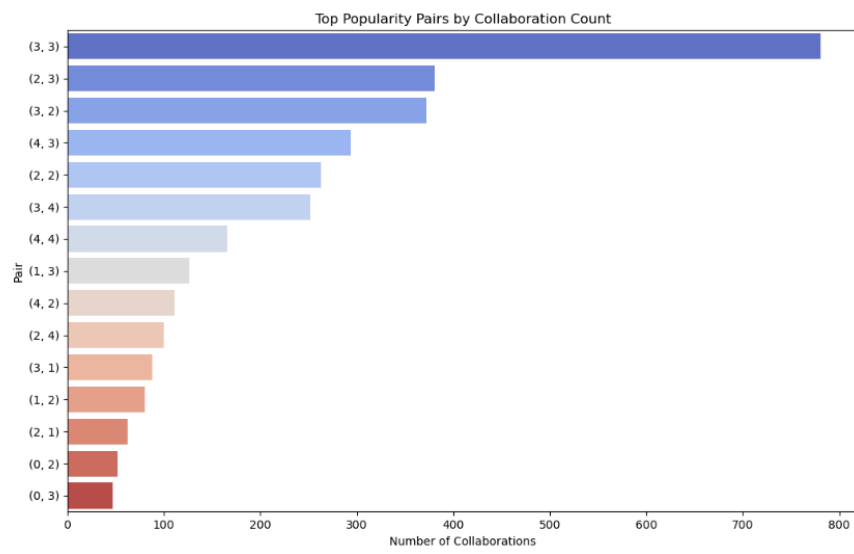


Figure 4: Top Popularity and Topic Pairs by Collaboration Count

The top popularity pairings—such as (3, 3), (2, 3), and (3, 2)—show that mid- to high-popularity creators collaborate most frequently. The presence of symmetric pairs (e.g., both (2, 3) and (3, 2)) indicates mutual collaboration across adjacent popularity bands.

Insight: Mid-tier influencers are most collaboration-active, while low-to-high tier collaborations are rare.

3.2 Topic-Based Collaborations

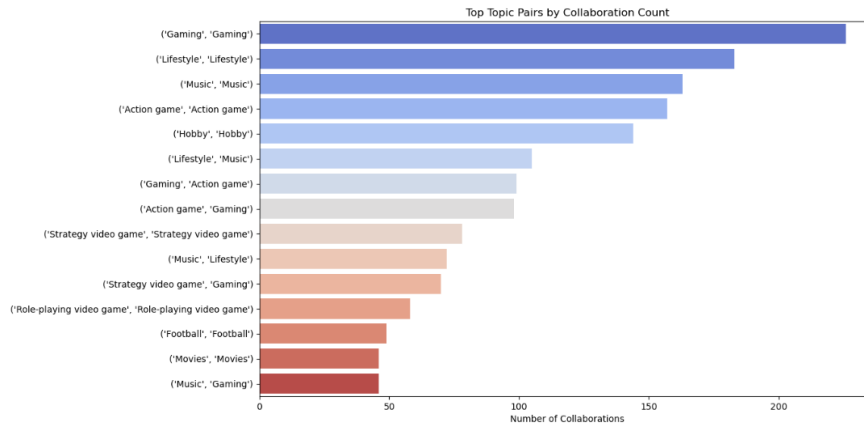


Figure 5: Top Language Pairs by Collaboration Count

Figure shows a dominance of intra-topic collaborations, particularly in:

- ('Gaming', 'Gaming')
- ('Lifestyle', 'Lifestyle')

- ('Music', 'Music')

Takeaway: Creators tend to collaborate within their topical niche, likely due to aligned audiences and shared content formats.

3.3 Category-Based Collaborations

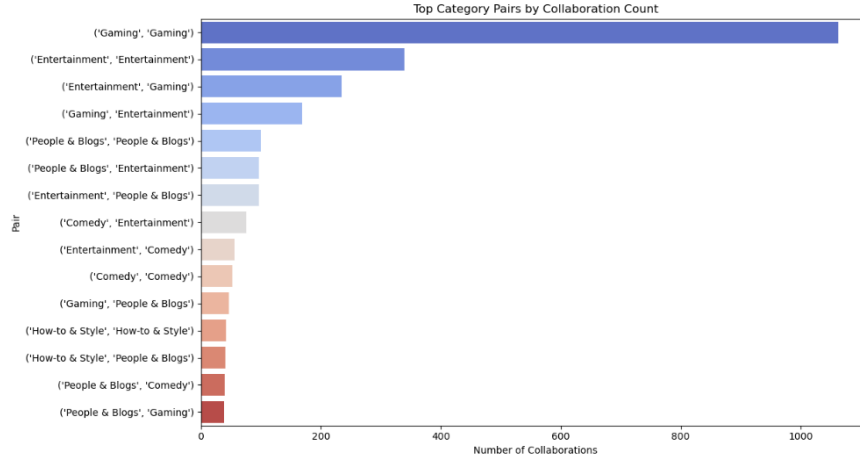


Figure 6: Top Category Pairs by Collaboration Count

We observe a strong clustering around intra-category collaborations—especially within the **Gaming**, **Entertainment**, and **People & Blogs** categories. Inter-category collaborations like ('Entertainment', 'Gaming') do occur but are relatively less frequent.

3.4 Language-Based Collaborations

Figure 7 displays the top 50 language pairings in collaborations. The key highlights include:

- **English (en)** is the most dominant language in collaborations, especially intra-language: (en & en) exceeds 1100 collaborations.
- Bilingual or closely related languages such as en & pt, pt & pt, en & es, and en & fr also feature prominently.
- **Intra-language collaborations dominate** the top list, but notable inter-language ties suggest some level of cross-lingual reach—possibly due to subtitles, bilingual creators, or culturally similar audiences.

Observation: Language acts as both a bridge and a barrier. While intra-language ties are strongest, multilingual collaborations are not negligible, especially among globally spoken languages (e.g., English, Spanish, Portuguese, Russian).

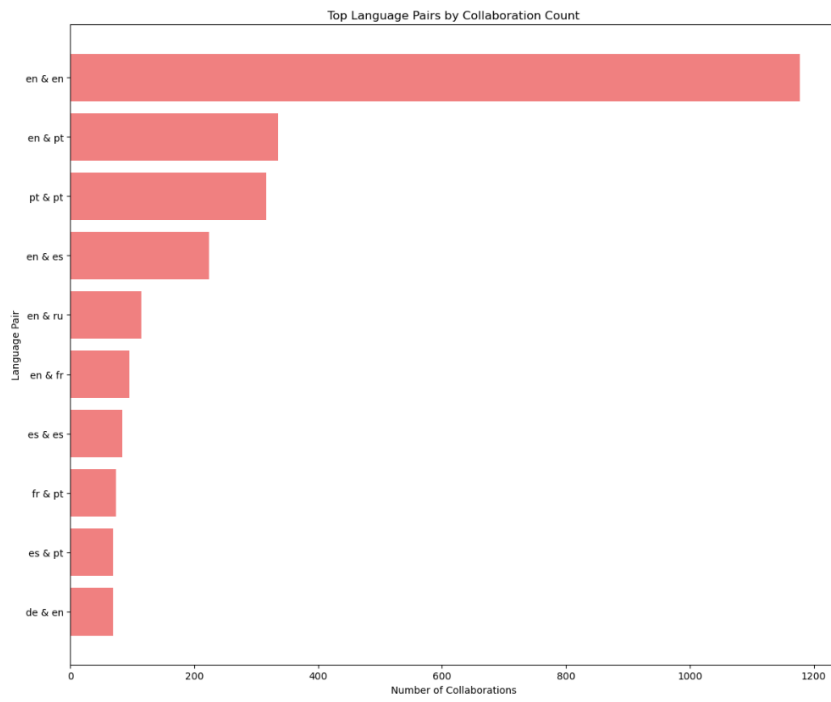


Figure 7: Top Language Pairs by Collaboration Count

3.5 Overall Conclusions

- **Popularity:** Most collaborations occur within mid- to high-popularity tiers, especially (2, 3) and (3, 3) pairs.
- **Topic:** Strong topical homophily—most creators collaborate within the same topic (e.g., Gaming or Lifestyle).
- **Category:** Intra-category links are dominant; Gaming and Entertainment lead in collaborative volume.
- **Language:** English leads all collaborations by a large margin; however, inter-language collaborations suggest potential for multilingual content strategies.

4 Impact Calculation and Analysis

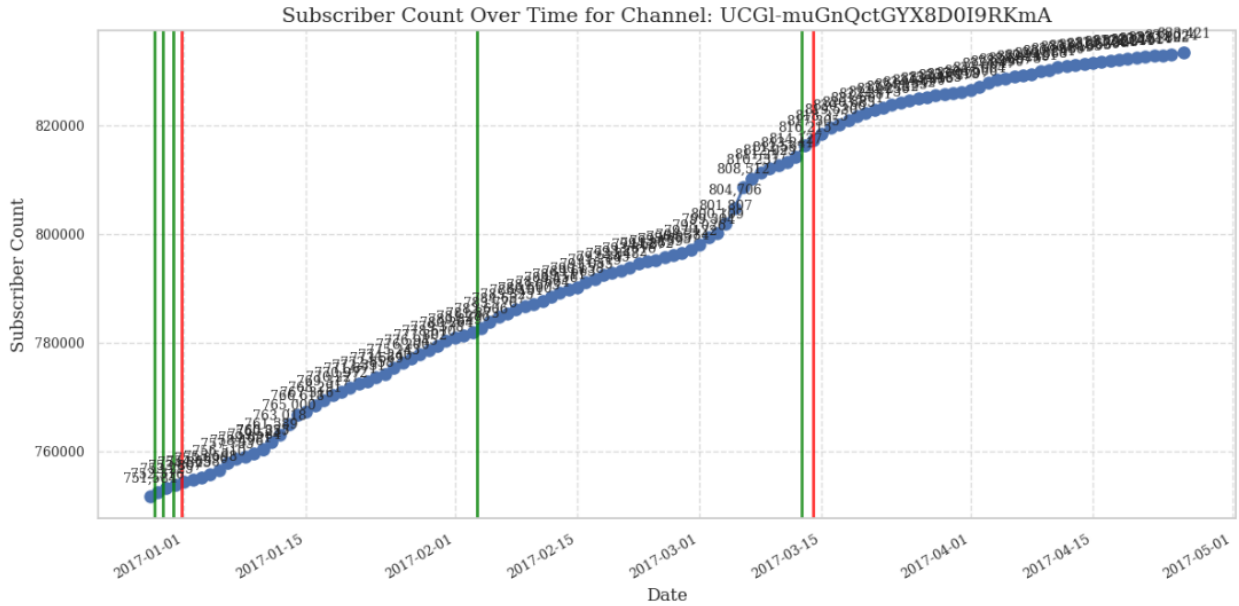
This report outlines the methodology to evaluate the influence of collaboration videos on the growth of YouTube subscribers. We employ a formula for calculating the impact of a collaboration by taking a non-collab window to find out the expected gradient and then comparing it with the gradient observed when the collab actually happens.

Data Preparation

We begin by reading and grouping data from a collaboration data set and a subscriber history data set. All of the collaboration information is given in the `df_collabs.txt`. The crawled data is given in the `df_channel_history_collabs.txt`. We take out all the channels from the `from_channel` column so that we can ensure that they are having at least one collaboration.

Exploration

We explored the data set to see the collab frequencies and gradient differences for different channels. Here the red lines denote the incoming collaborations, green lines denote the outgoing collaborations, and the blue graph denotes the subscriber count.



where $\hat{m}_{\text{baseline}} = \frac{d\hat{S}}{dt}$ is the predicted gradient from linear regression on the baseline period, and $m_{\text{collab}} = \frac{dS}{dt}$ is the observed gradient during the collaboration period.

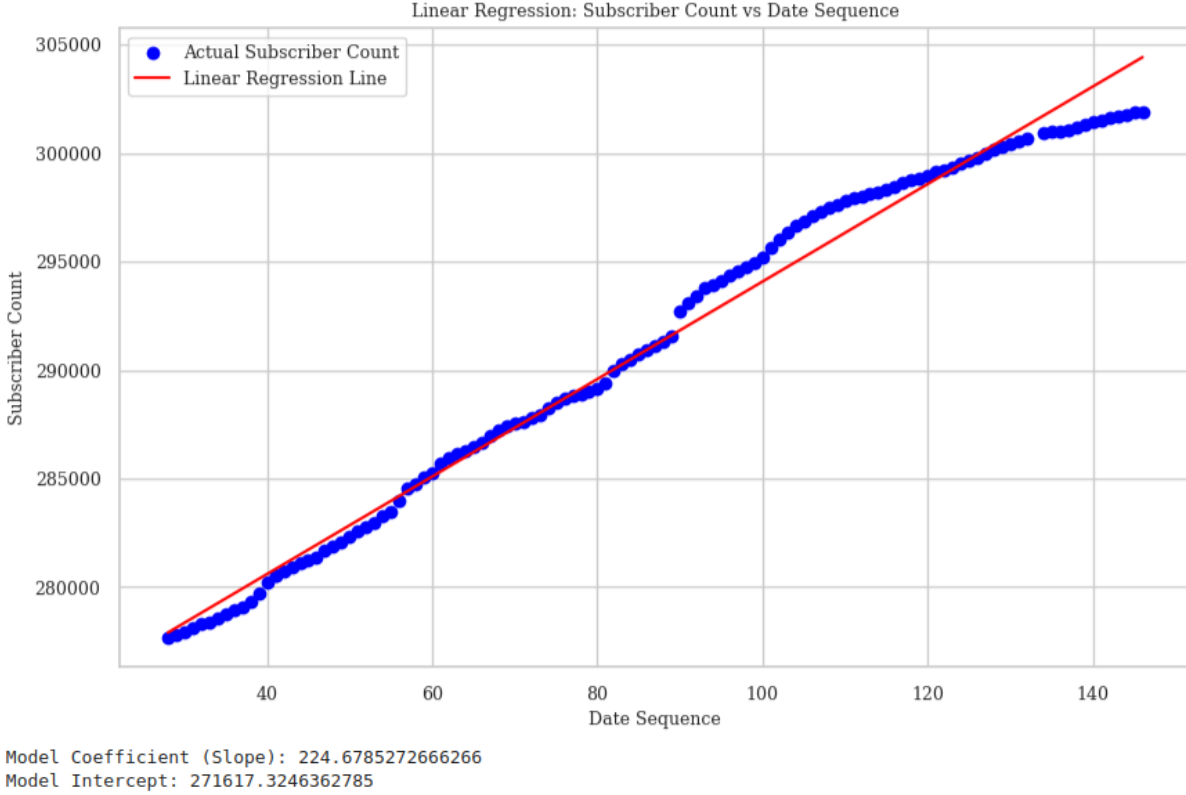


Figure 9: Example of Linear Regression on one channel

- **Method 2: Seven days before and after** We picked the gradient from -7 days to date of collab and from collab date to +7 days. We then calculated their difference. This gave us gradient differences which seemed a bit vague, because the gradient at -7 days could have been due to some previous collab as well.

$$\text{Impact} = (S_{d+7} - S_d) - (S_d - S_{d-7})$$

where S_d is the subscriber count on day d . This method compares the change before and after the collab date but may include interference from earlier events.

- **Method 3: First Non-Collab Window** We picked the first non-collab window of 5 days to see the baseline gradient for a channel. Then we took the gradient of 5 days after the collaboration took place and took the difference in those values to calculate the impact. This method solved the problem which was in method 2. We then normalized the values using the slope obtained with the linear regression

$$\text{Impact} = \frac{(S_{d+5} - S_d) - (S_b - S_a)}{m}$$

where S_a to S_b represent subscriber counts in the initial 5-day non-collab window, S_d is the subscriber count on the collab day, and m is the linear regression slope (baseline growth rate).

Analysis

We then calculated the impact for other channels as well. They are present in the **channel_and_max_impact_colla** and using this we tried to find out if the frequency of collaborations has any relation with the max impact with a particular topic.

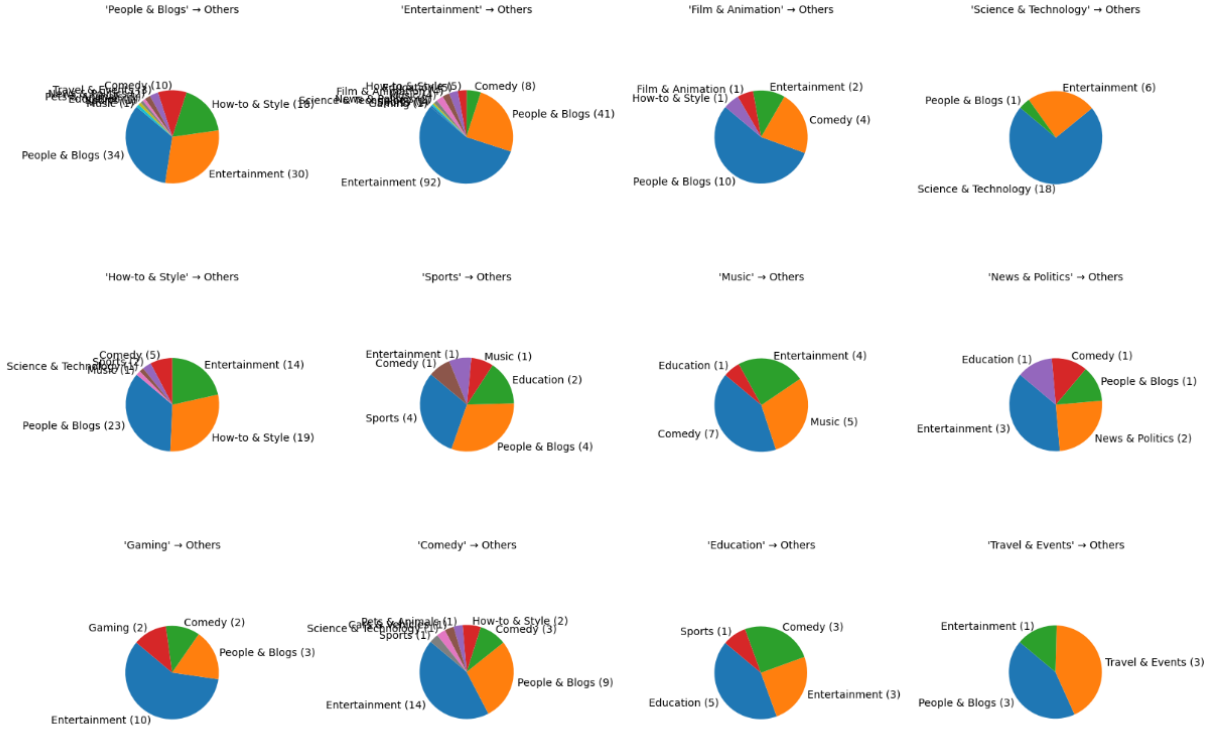
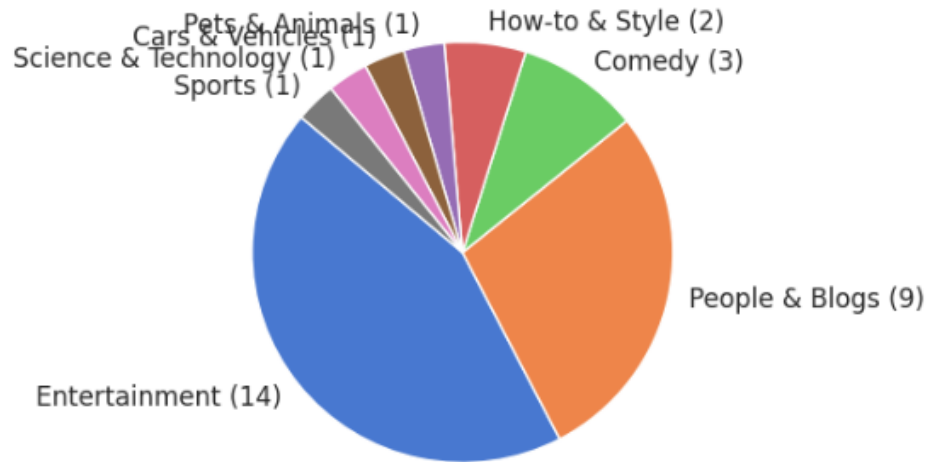


Figure 10: Analysis on different topics

Above is the analysis for 12 different topics on which the channels are segregated. For one particular, we have tried to observe which topics are having the max frequency of collaborations and which topics are having the max impact, and are these the same with each other.

Collaborations of 'Comedy' Channels with Other Categories



Top 10 'to_category' by max_impact for from_category = 'Comedy':

to_category	
Comedy	1236.600167
Entertainment	586.948258
People & Blogs	405.666455
How-to & Style	351.345732

Figure 11: Analysis on different topics

Above is the analysis on the "comedy" topic with other topics, here we can see that even though the collaboration frequency is maximum with "Entertainment" and "People and Blogs", the top impact is arising from collaborating with a comedy channel.

Observations from this study

This study shows that channels don't always get the most benefit from the ones they collaborate with the most. While many channels often work with popular categories like "Entertainment" or "People and Blogs," the biggest growth or impact sometimes comes from working with different, less frequent partners. This means that choosing the right partner matters more than just working with someone often. Finding the best match in terms of content can lead to better results, which we have observed comes from collaborating with a channel of the same nature.

5 Clustering Analysis

Here, we have used **Spectral Clustering**, a graph-based technique effective for detecting complex, non-linear cluster structures. We chose Spectral Clustering over Louvain because **Louvain cannot be applied to directed graphs**, which are present in this analysis. Using this approach, we have formed **40 distinct clusters** from the dataset.

We began with an initial graph consisting of **3625 nodes** and **3475 edges**. After removing **1170 isolated nodes**, the final graph used for clustering had **2455 nodes** and **3475 edges**. Spectral Clustering was then applied to this graph to identify meaningful clusters.

Cluster Statistics :

Cluster	Node Count	% of Graph
0	834	33.97%
1	54	2.20%
2	20	0.81%
3	13	0.53%
4	48	1.96%
5	35	1.43%
6	155	6.31%
7	23	0.94%
8	15	0.61%
9	46	1.87%
10	23	0.94%
11	19	0.77%
12	40	1.63%
13	14	0.57%
14	39	1.59%
15	14	0.57%
16	35	1.43%
17	6	0.24%
18	18	0.73%
19	30	1.22%

Cluster	Node Count	% of Graph
20	26	1.06%
21	52	2.12%
22	8	0.33%
23	10	0.41%
24	38	1.55%
25	15	0.61%
26	13	0.53%
27	431	17.56%
28	8	0.33%
29	22	0.90%
30	61	2.48%
31	59	2.40%
32	43	1.75%
33	11	0.45%
34	4	0.16%
35	4	0.16%
36	82	3.34%
37	37	1.51%
38	10	0.41%
39	40	1.63%

Cluster Distribution Graph:

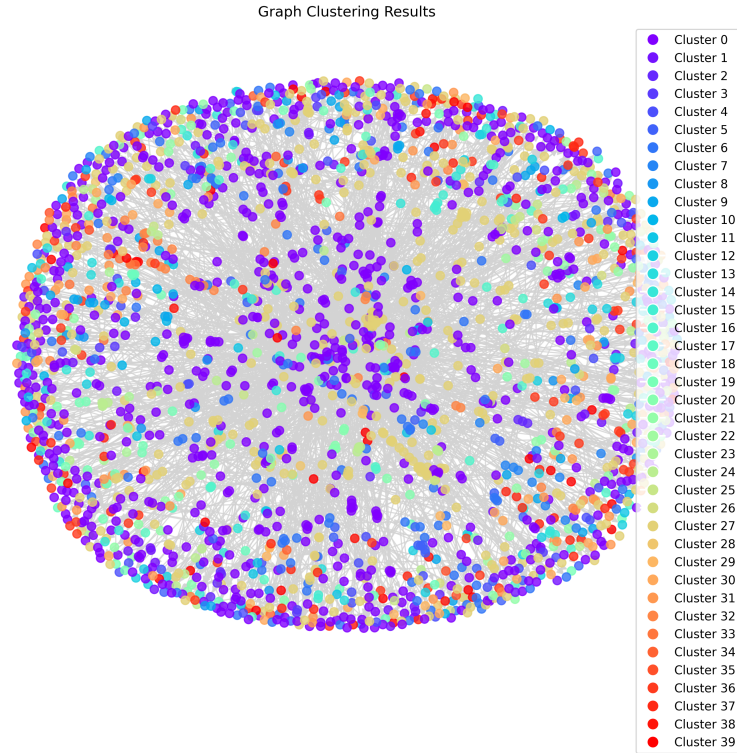


Figure 12: Distribution of nodes across the 40 clusters.

Cluster Density (Internal Connectivity):

The internal connectivity of each cluster was measured using density values, which reflect how tightly the nodes are interconnected within each cluster. The density vary significantly across clusters:

- **Cluster 0**, which contains the largest number of nodes (834), has a low density of **0.0043**, indicating it is a sparsely connected cluster.
- In contrast, smaller clusters such as **Cluster 22 (density = 0.3571)**, **Cluster 28 (0.3929)**, and **Clusters 34 and 35 (0.3333)** exhibit significantly higher density values. These high density suggest strong internal connectivity and tightly-knit node groups.

Cluster Density Table:

Cluster	Density	Cluster	Density	Cluster	Density
0	0.0043	1	0.0224	2	0.1211
3	0.1154	4	0.0239	5	0.0739
6	0.0184	7	0.1660	8	0.1524
9	0.0396	10	0.1067	11	0.1287
12	0.0359	13	0.1538	14	0.0445
15	0.1538	16	0.0689	17	0.2000
18	0.0915	19	0.0690	20	0.0892
21	0.0264	22	0.3571	23	0.2222
24	0.0583	25	0.1810	26	0.1026
27	0.0064	28	0.3929	29	0.1039
30	0.0246	31	0.0187	32	0.0565
33	0.2364	34	0.3333	35	0.3333
36	0.0163	37	0.0315	38	0.1778
39	0.0551				

6 Cluster Analysis

We analyzed clusters to identify collaborative behaviors, dominant languages, and popularity among YouTube channels.

6.1 Language Distribution by Cluster

Cluster	Languages	Cluster	Languages	Cluster	Languages
0	en (43%), es (13%), de (4%)	1	en (59%), ar (4%)	2	en (18%)
3	en (31%), es (23%)	4	en (68%)	5	en (62%)
6	en (59%), es (14%)	7	en (74%), de (9%)	8	en (36%), es (7%)
9	en (37%), es (7%)	10	de (45%), en (23%)	11	en (32%)
12	en (65%), de (14%), ar (5%)	13	en (31%), de (8%)	14	en (40%), ar (9%)
15	en (14%), de (14%)	16	en (54%), ro (3%)	17	en (100%)
18	en (57%)	19	en (31%)	20	en (36%)
21	en (62%)	22	en (60%)	23	en (84%)
24	en (32%)	25	en (53%)	26	en (47%)
27	en (36%)	28	en (82%)	29	en (45%)
30	en (57%)	31	en (65%)	32	en (44%)
33	en (59%)	34	en (100%)	35	en (100%)
36	en (46%)	37	en (47%)	38	en (62%)
39	en (41%)				

6.2 Average Collaborations per Cluster

Cluster	Avg. Collaborations	Cluster	Avg. Collaborations	Cluster	Avg. Collaborations
0	3.77	1	1.19	2	2.85
3	1.38	4	1.12	5	2.77
6	3.08	7	4.13	8	2.67
9	1.78	10	2.78	11	2.68
12	1.40	13	2.07	14	1.69
15	2.07	16	3.26	17	1.00
18	1.56	19	2.60	20	2.62
21	1.35	22	2.62	23	3.20
24	2.16	25	3.53	26	1.23
27	3.03	28	2.75	29	2.36
30	1.48	31	1.08	32	2.79
33	3.00	34	1.00	35	1.00
36	1.32	37	1.14	38	1.60
39	2.48				

6.3 Average Popularity per Cluster

Cluster	Avg. Popularity	Cluster	Avg. Popularity	Cluster	Avg. Popularity
0	2.44	1	2.50	2	2.50
3	2.62	4	2.65	5	2.74
6	2.60	7	2.52	8	3.07
9	2.83	10	2.87	11	2.84
12	2.35	13	2.57	14	2.64
15	3.07	16	2.43	17	1.83
18	2.28	19	2.70	20	3.35
21	2.58	22	2.75	23	2.90
24	2.24	25	3.07	26	2.54
27	2.87	28	2.25	29	2.50
30	2.28	31	2.34	32	2.65
33	2.82	34	2.50	35	3.00
36	2.51	37	2.41	38	2.30
39	2.35				

Conclusion: We conclude that there are no specific feature that divides channels from each other in clusters

6.4 Centrality-Based Collaboration Analysis

To further understand the collaboration patterns within the YouTube channel network, we analyzed the centrality of nodes (channels) and their collaboration behaviors. Centrality metrics, such as Degree Centrality, Betweenness Centrality, Closeness Centrality, Eigenvector Centrality, and PageRank, were used to quantify the influence and connectivity of nodes.

Results of Centrality Analysis: The analysis revealed distinct collaboration patterns based on centrality metrics. The results are summarized in the table below, showing the percentage of collaboration types and the inferred behavior for each centrality metric.

Metric	High → High	High → Low	Low → High	Low → Low	Inference
Degree Centrality	37.81%	24.52%	14.99%	22.68%	High-centrality nodes collaborate broadly
Betweenness Centrality	30.39%	26.73%	13.18%	29.70%	High-centrality nodes collaborate broadly
Closeness Centrality	39.94%	0.26%	19.97%	39.83%	Mixed collaboration behavior
Eigenvector Centrality	40.03%	0.63%	19.34%	40.00%	Mixed collaboration behavior
PageRank	29.47%	9.35%	33.27%	27.91%	Low-centrality nodes seek high-centrality nodes

Insight: The analysis shows that for Degree and Betweenness Centrality, high-centrality nodes tend to collaborate broadly, including with low-centrality nodes, suggesting they act as hubs in the network. For PageRank, low-centrality nodes actively seek collaborations with high-centrality nodes, indicating a desire to connect with influential channels. Closeness and Eigenvector Centrality show mixed behaviors, with no single dominant collaboration pattern.

6.5 Cluster Plot Visualizations

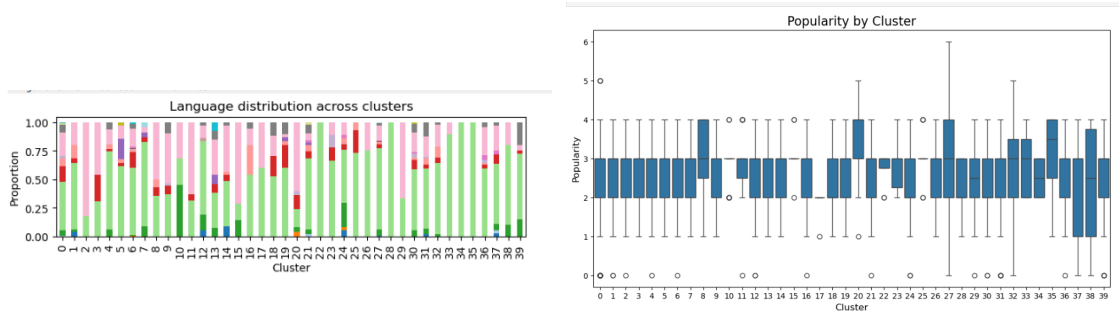


Figure 13: Language distribution (left) and average popularity (right) by cluster

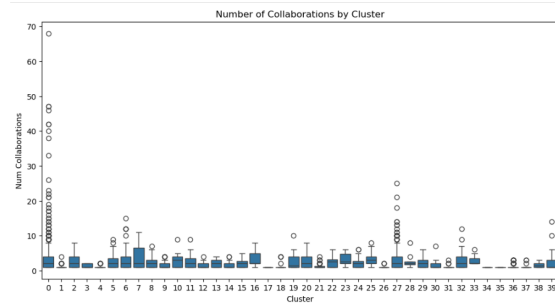


Figure 14: Average collaborations by cluster

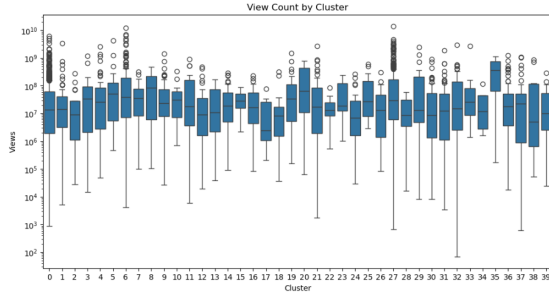


Figure 15: View Count by Cluster

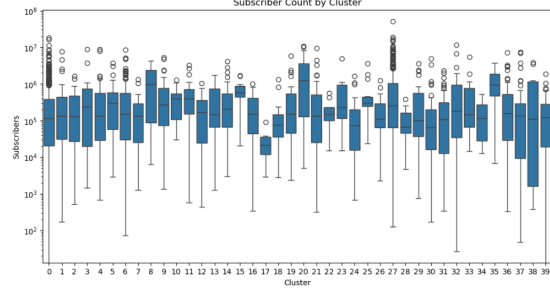


Figure 16: Subscriber Count by Cluster

7 Intra-Cluster Collaborations

We analyzed whether the collaborating channels belong to the same cluster. The results show that:

Total Collaborations: 3475

Collaborations within Same Cluster: 3217

Percentage of Intra-Cluster Collaborations: 92.58%

Insight: A high percentage (92.58%) of collaborations occur within the same cluster, indicating that channels prefer collaborating with similar or closely related entities.

Top 10 Channels by Number of Collaborations:

Channel ID	Cluster	Num Collaborations	Category	Topic
UC-Tvz0FJd6xRvVDw ₄ dc _G Q	0	68	Entertainment	Strategy video game
UCzIzWHfn7wsS _V K0vQH _V 8E _w	0	47	Gaming	Sports game
UCbTRgghKvt-I _V 8CUtCP _{Yr} A	0	47	Entertainment	Gaming
UCAcziR0Lap _N IgQcwH7G29A	0	46	Gaming	Strategy video game
UCZjCiuXIP9ydZ4E6lxq1rIw	0	42	Gaming	Action game
UCm4WIdrdOOSbht-NKQ0uTeg	0	42	Gaming	Action game
UCUouM5ZRqZoFNzUvpDeYNGQ	0	40	Gaming	Strategy video game
UC-iNzkiXFbRr _l - lq9MVy7w	0	38	Gaming	Gaming
UC _U far - P7FwrYAyznm96bA	0	33	Gaming	Action game
UCwK3b21gUfX4 _h a6uOLRrlA	0	26	Gaming	Football

8 Channel Collaboration Analysis

We analyze the collaboration behavior of individual channels within a specific cluster. Here, we present an example analysis for the channel UC-Tvz0FJd6xRvVDw₄dc_GQ, which belongs to Cluster 0.

Channel: UC-Tvz0FJd6xRvVDw₄dc_GQ

Belongs to Cluster: 0

Intra-Cluster Collaboration: This channel predominantly collaborates within its own cluster, indicating strong intra-cluster cohesion. The majority of this channel's collaborations (94%) are within Cluster 0. This suggests that the channel is part of a tightly connected network, collaborating frequently with similar creators.

Bridge Nodes to Other Clusters: The remaining collaborations (6%) are with channels in Clusters 10, 16, and 27. These few collaborations may serve as bridges, facilitating cross-cluster interactions. These bridge collaborations help spread influence and foster new relationships, potentially increasing the channel's reach beyond its primary community.

Breakdown of Collaborations:

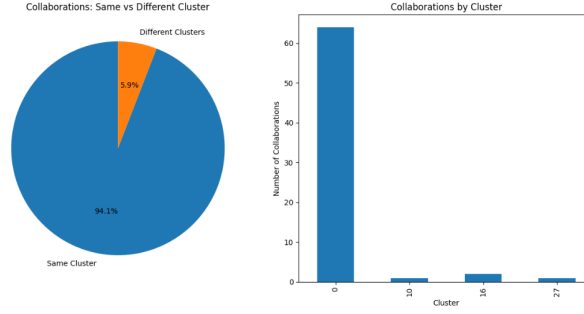


Figure 17

Cluster	Collaborations	Percentage
Cluster 0	64	94.12%
Cluster 10	1	1.47%
Cluster 16	2	2.94%
Cluster 27	1	1.47%

Conclusion: UC-Tvz0FJd6xRvVDw_4dc_GQ serves as a key node in Cluster 0, with the majority of its collaborations within the cluster. Channels from other clusters (like 10, 16, and 27) can benefit from collaborating with this channel to expand their network and visibility, as it acts as a bridge for cross-cluster interactions.

General Insight: This strategy can be applied to other channels as well. Channels in Cluster 0 can serve as valuable collaboration points for channels from different clusters to increase their reach and influence.

9 Top 10% Nodes by Centrality

Category	Count	Percentage (%)
Gaming	194	49.62
Entertainment	89	22.76
People & Blogs	31	7.93
Comedy	23	5.88
Film & Animation	12	3.07
Education	11	2.81
How-to & Style	10	2.56
Music	9	2.30
Sports	4	1.02
News & Politics	4	1.02
Science & Technology	3	0.77
Travel & Events	1	0.26

Table 1: Top 10% Central Nodes by Category (Count and Percentage)

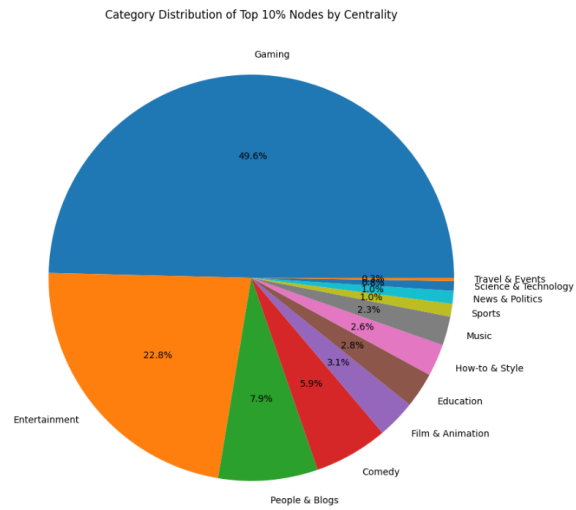


Figure 18: Category Distribution of Top 10% nodes

Conclusion

The results indicate that the **Gaming** category dominates in terms of both the number and proportion of highly central channels, accounting for nearly half (49.62%) of the top 10% most central nodes. This is followed by **Entertainment** with 22.76%. These findings suggest a highly influential role played by gaming and entertainment content in the structure of the YouTube channel network.